



TOOL CATALOG

BR1163-04

MS004003/06E

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

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PREFACE

This tool catalog comprises tools and devices required to carry out maintenance and assembly work. The safety instructions in the relevant operating and maintenance manuals must be observed.

Tool catalog structure

The content is subdivided into the following categories:

- general tools
- qualification level-dependent tools
- workshop equipment
- measuring devices

Within these categories, the tools are arranged by function groups. An overview over these function groups is provided on the following pages.

The below symbols are used for the combination of several tools:



tool kit



individual tool within a tool kit



tool which must be used in combination with other tools



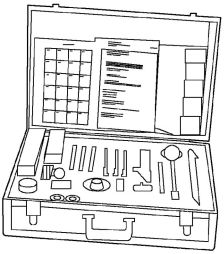
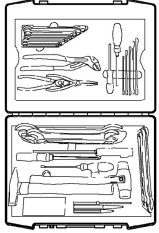
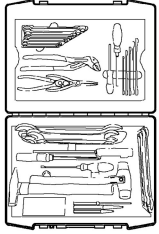
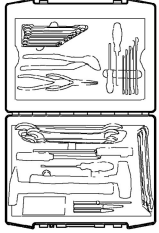
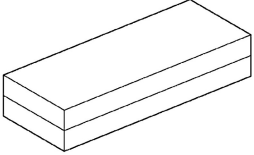
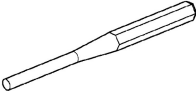
spare tool with position number from illustration

Note: The illustration of the tools may deviate to a minor extent.
When ordering tools, the following information must be provided:

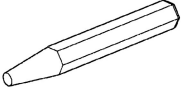
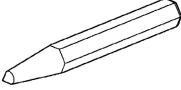
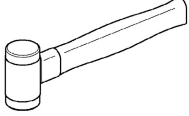
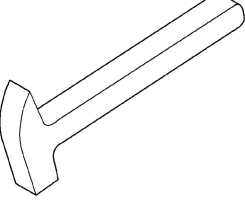
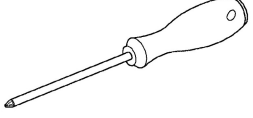
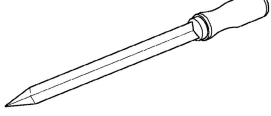
Engine type		Designation	
Part No.		Engine No.	
Order No.		Tool catalog No.	
Quantity			

To avoid transmission errors, please confirm any orders placed over the telephone in writing.

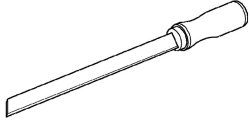
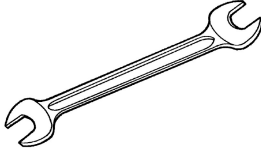
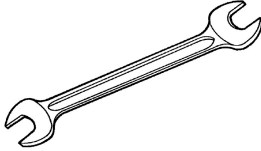
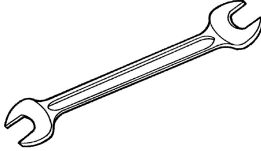
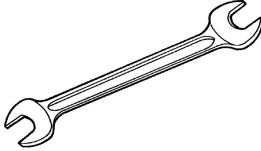
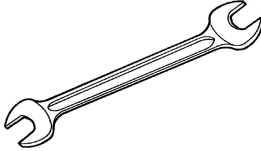
TOOL GENERAL TOOLS

SYM	ORDER NO.	DESIGNATION USAGE	QTY.	IMAGE
	F6798833	TEST KIT to analyze coolant and engine oil samples	1	
	F30454308	TOOL KIT general tool kit	1	
	F30454308/01	TOOL BOX	1	
	F30454308/02	COMPARTMENTS	1	
	F30454308/03	SHARP.STONE, FLAT	1	
	F30454308/04	PIN PUNCH 3x150	1	

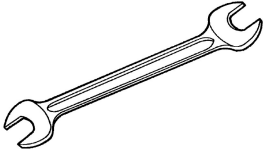
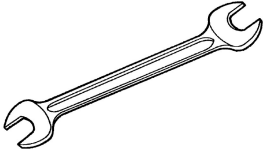
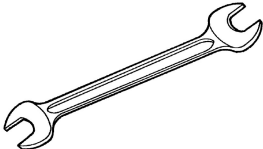
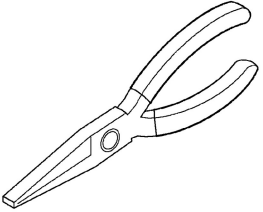
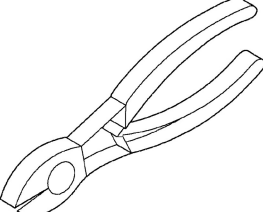
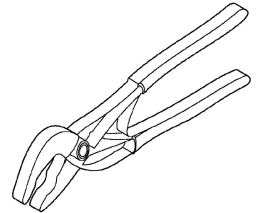
TOOL GENERAL TOOLS

SYM	ORDER NO.	DESIGNATION USAGE	QTY.	IMAGE
	F30454308/05	DRIFT PUNCH 5x120	1	
	F30454308/06	CENTER PUNCH	1	
	F30454308/07	HAMMER Synthetic material	1	
	F30454308/08	HAMMER 500gr.	1	
	F30454308/09	SCREWDRIVER 1,2x7x125	1	
	F30454308/10	THREE-SQUARE HOLLOW SCRAPER	1	

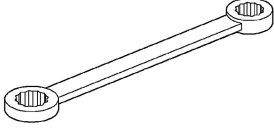
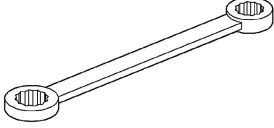
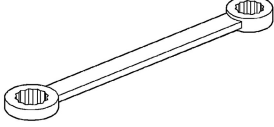
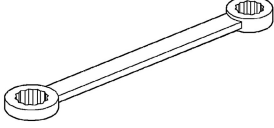
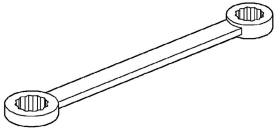
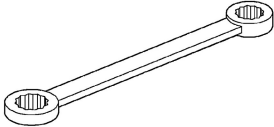
TOOL GENERAL TOOLS

SYM	ORDER NO.	DESIGNATION USAGE	QTY.	IMAGE
	F30454308/11	FLAT SCRAPER	1	
	F30454308/12	OPEN-END WRENCH wrench size 30x32	1	
	F30454308/13	OPEN-END WRENCH wrench size 24x27	1	
	F30454308/14	OPEN-END WRENCH wrench size 30x32	1	
	F30454308/15	OPEN-END WRENCH wrench size 36x41	1	
	F30454308/16	OPEN-END WRENCH wrench size 17x19	1	

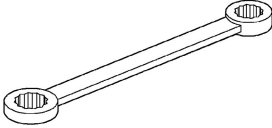
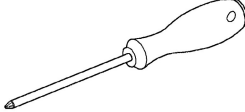
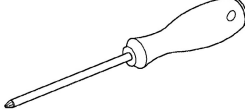
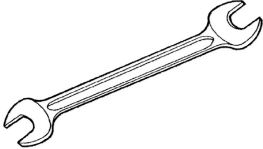
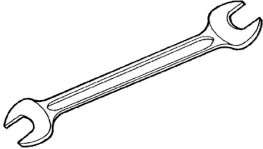
TOOL GENERAL TOOLS

SYM	ORDER NO.	DESIGNATION USAGE	QTY.	IMAGE
	F30454308/17	OPEN-END WRENCH wrench size 20x22	1	
	F30454308/18	OPEN-END WRENCH wrench size 22x24	1	
	F30454308/19	OPEN-END WRENCH wrench size 24x27	1	
	F30454308/20	COMBINATION PLIERS	1	
	F30454308/21	DIAGONAL CUTTING PLIERS	1	
	F30454308/22	WATER PUMP PLIERS	1	

TOOL GENERAL TOOLS

SYM	ORDER NO.	DESIGNATION USAGE	QTY.	IMAGE
	F30454308/23	DOUBLE-HEAD BOX WRENCH wrench size 8x9	1	
	F30454308/24	DOUBLE-HEAD BOX WRENCH wrench size 10x11	1	
	F30454308/25	DOUBLE-HEAD BOX WRENCH wrench size 12x13	1	
	F30454308/26	DOUBLE-HEAD BOX WRENCH wrench size 13x15	1	
	F30454308/27	DOUBLE-HEAD BOX WRENCH wrench size 16x17	1	
	F30454308/28	DOUBLE-HEAD BOX WRENCH wrench size 18x19	1	

TOOL GENERAL TOOLS

SYM	ORDER NO.	DESIGNATION USAGE	QTY.	IMAGE
	F30454308/29	DOUBLE-HEAD BOX WRENCH wrench size 20x22	1	
	F30454308/30	SCREWDRIVER 0,5x3x80	1	
	F30454308/31	SCREWDRIVER 1,5x0,5x100	1	
	F30454308/32	OPEN-END WRENCH wrench size 6x7	1	
	F30454308/33	OPEN-END WRENCH wrench size 8x10	1	
	F30454308/34	OPEN-END WRENCH wrench size 10x11	1	